typesense

Personalizing Search Results

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Personalization, in the context of building search experiences, refers to the process of ranking search results differently for searchers based on unique characteristics we might know about them.

Personalization mechanisms come in two flavors:

1. **User-group level personalization -** where results are ranked uniquely for each *group* of users, depending on that group's aggregated characteristics (subscription plan they're on, user role, high LTV customers, etc) and/or past behaviors common to users in that group.
2. **User-level personalization -** where results are ranked uniquely for each specific user depending on their specific past behavior (past searches, past purchases, past clicks, etc).

User-group level personalization

Let's first see how we can implement **user-group level personalization** using Typesense's popularity-based ranking feature.

This is a much simpler and easier-to-implement approach compared to user-level personalization which requires the use of Machine Learning.

Let's use an example of a hardware tools e-commerce store, where users can pay to join one of three paid subscription plans:

1. Hobbyist plan
2. Advanced plan
3. Professional plan

Let's say we want to show users a different set of products when they search for common terms, depending on the plan they are on.

For eg, if a user on the Hobbyist plan searches for the term "screwdriver", we want to show them small screwdriver sets suitable for Hobbyists. But if a user on the Professional plan searches for the

Here's how we would implement this with Typesense.

Let's say we have the following two product records:

```json
[
  {
    "name": "3-piece Screwdriver Set",
    "price": 10.0,
    "category": "Tools"
  },
  {
    "name": "120-piece Power Screwdriver Set",
    "price": 250.0,
    "category": "Tools"
  }
]
```

We can add plan-specific popularity scores to each record based on heuristics, or any other information we might already know about each group like the following:

```json
[
  {
    "name": "3-piece Screwdriver Set",
    "price": 10.0,
    "category": "Tools",
    "hobbyist_plan_popularity_score": 100,
    "advanced_plan_popularity_score": 50,
    "professional_plan_popularity_score": 10
  },
  {
    "name": "120-piece Power Screwdriver Set",
    "price": 250.0,
    "category": "Tools",
    "hobbyist_plan_popularity_score": 10,
    "advanced_plan_popularity_score": 50,
    "professional_plan_popularity_score": 100
  }
]
```

Now let's say a user in the Hobbyist Plan logs in, in order to show them results personalized to their user group, we would add this `sort_by` parameter to the search query:

```
sort_by=_text_match(buckets: 10):desc,hobbyist_plan_popularity_score:desc
```

This would cause Typesense to interweave text relevance with popularity score for that user's plan (Hobbyist) and show the "3-piece Screwdriver Set" first.

Similarly, if a user in the Professional Plan logs in, we would add this `sort_by` parameter to the search query:

```
sort_by=_text_match(buckets: 10):desc,professional_plan_popularity_score:desc
```

This would cause Typesense to interweave text relevance with popularity score for that user's plan (Professional) and show the "120-piece Power Screwdriver Set" first.

Use-cases

You could use this concept to show personalized results to different users based on:

- Customer Life-Time Value buckets (high, medium, low spenders)
- Tenure of their account (accounts older than 5 years, 2 years and new accounts)
- Age group (if you've collected this information already)
- Gender (for eg, you could rank products relevant to women higher if the user has indicated that they are female).
- Geographic location (for eg, you could rank products differently for your customers in the US vs Europe).

User-level personalization

Typesense's Vector Query Re-Ranking feature allows you to bring embeddings from an ML model (that takes users' past behaviors as input), index them in Typesense and then use that to re-rank results personalized to each user's past behavior.

Instead of sending the embedding to the `vector_query` parameter (which generates recommendations), you'd send it to the `sort_by` parameter like below (which re-ranks results in a personalized way):

```sh
curl "${TYPESENSE_URL}/multi_search" \
        -X POST \
        -H "Content-Type: application/json" \
        -H "X-TYPESENSE-API-KEY: ${TYPESENSE_API_KEY}" \
        -d '{
          "searches": [
            {
              "collection": "products",
              "q": "shoes",
              "query_by": "title,brand,category",
              "sort_by": "_text_match:desc,_vector_query(embedding:(['${embedding%?}
            }
          ]
        }' | jq '.results[0].hits[]'
```

Typesense will first do a keyword search for `shoes`, and then use the embeddings generated by the ML model for this particular's user's session behavior to re-rank the results such that the records closest to this user's preferences are ranked higher.

This feature is available as of `v0.25.2` of Typesense.

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